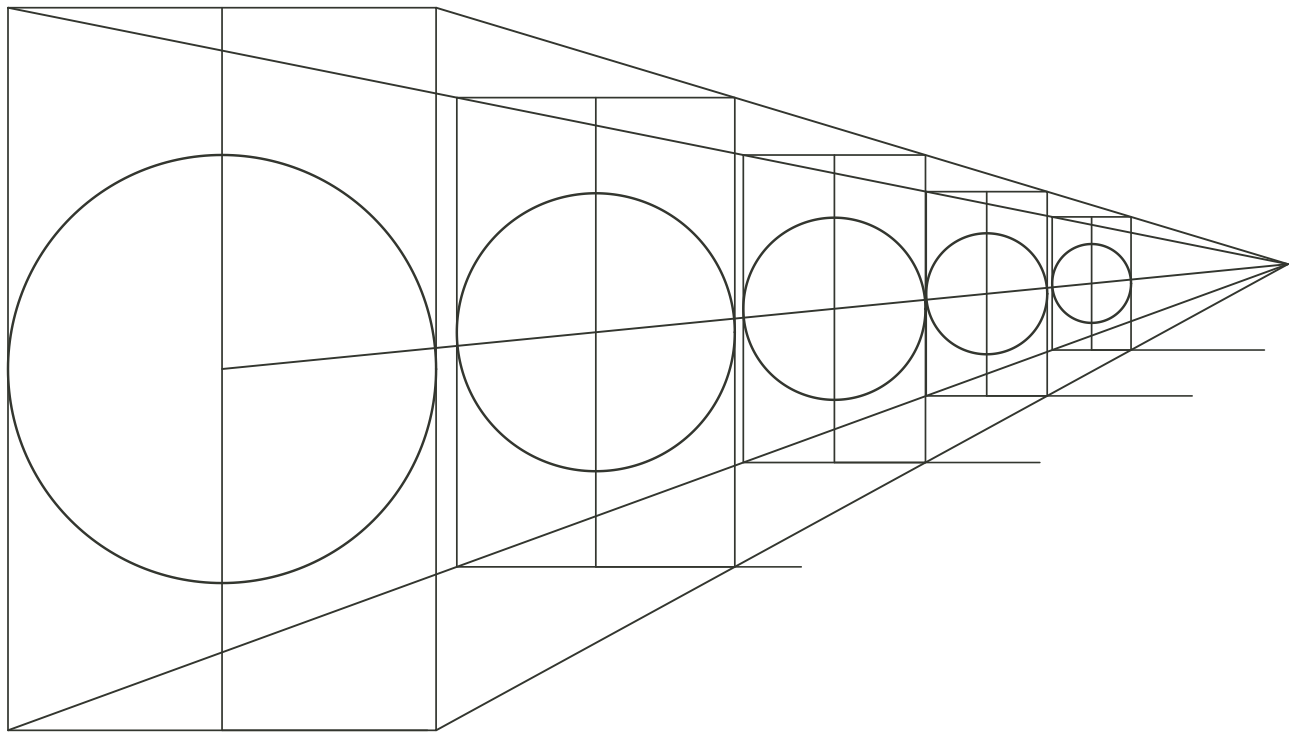


# Safety–Capability Coupling Program

A mathematical account of the intended mechanism,  
established results, and remaining construction requirements.



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**Version 1.0**

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Evidence through LN-239. A synthesis of an active research program, prepared as a precursor to a formal paper. The record establishes neither a working intrinsic SCC mechanism nor a general impossibility theorem.

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## Executive summary

Safety–Capability Coupling (SCC) asks whether the computations implementing a model’s alignment can be made indispensable to its useful cognition. The intended consequence is destructive: removing the protected alignment function should cause catastrophic loss of the individual model’s cognitive function. The present experiments establish only restricted analogues on specified tasks. Neither a refusal failure nor a decrease in benchmark accuracy is that endpoint.

The research has identified three claims that require separate arguments:

1. **Conditional functional coupling:** useful cognition cannot coexist with genuine loss of a specified protected function.
2. **Behavioral enforcement:** an attacker cannot obtain prohibited behavior while retaining useful cognition, including by keeping the protected function and disregarding its judgment.
3. **Durability:** allowed preservation and repair cannot recover the capability after the relevant destructive event.

These claims have different counterexamples. Restoring both cognition and protection can defeat durability without defeating conditional coupling. Conversely, an intact-judgment override can defeat enforcement without demonstrating that the judgment was removed. This separation is essential to interpreting the record.

The strongest established conclusions concern explicit boundaries. Copying, alternate execution and substitution defeat constructions whose contracts admit the required operations. An exact one-probe storage separation disappears under a two-probe interface with one additional summary bit. Information-theoretic results can force loss of specified prediction functions while leaving generic learning intact. Finite recovery experiments show why reconstruction cost must include retained programs, observations, live state and permitted copies.

There are also legitimate positive reference results. Non-malleable encodings support conditional conclusions under specified tampering and commit assumptions. Classical self-erasing functions establish consumption under total-memory and communication restrictions. Quantum key leasing and private circuits establish different guarantees under their own cryptographic or physical contracts. None currently provides the missing intrinsic learned SCC construction.

The unresolved requirement is a concrete connection between an independently meaningful alignment computation and indispensable useful computation that survives the admitted alternate implementations. A candidate must also explain the behavior and repair properties it claims. A resource restriction helps only when its continued validity after editing is justified.

**Current assessment.** The record supports scoped positive and negative results, a better measurement framework, and explicit construction requirements. It does not support scaling the existing transform or encoding constructions into a successful neural mechanism merely by increasing their size. It also does not support treating an unsuccessful candidate search as a universal impossibility proof.

**Reading guide.** Sections 1–3 define the problem. Sections 4–10 develop the mathematical results and their limits. Sections 11–13 describe the remaining mechanism requirements. Appendices supply proof details, an evidence map and references. Internal source references identify dated labnotes; external references identify published or publicly available research. Numerical results are reported from the preserved record, not newly reproduced for this whitepaper.

## 1. Intended mechanism and scope

### 1.1 The construction objective

Let a model carry out useful reasoning and a protected alignment computation. The objective is to make the latter indispensable to the former in the individual model: removing the protected function should destroy the computations needed for useful cognition. Shared parameters are insufficient. The claim concerns what a modified model can still compute, including through permitted alternative representations and execution procedures.

The target is stronger than making modification inconvenient. It is also stronger than enforcing a permission check through an external trusted service. Such engineering may be valuable, but its trust boundary is part of the explanation and cannot be attributed to an intrinsic learned dependency.

The work proceeds through small, explicitly bounded constructions because the complete cognitive endpoint is not yet operationally defined by the existing task suite. A bounded result must identify the task abilities that were present, the protected function removed, and the severity of subsequent loss. A model falling below a 95% utility-retention threshold has failed a retention test; it has not thereby lost all cognition.

### 1.2 Four distinctions that govern interpretation

**Representation versus function.** Replacing stored coordinates, changing a readout, or destroying a particular implementation need not remove the function. An alternative decoder can preserve the same judgment.

**Information versus computation.** Erasing knowledge of an old task instance does not establish destruction of the algorithm that can learn from new observations. Conversely, information can remain present while access or computational limits prevent its timely use.

**Judgment versus action.** A system may correctly determine that an action is prohibited and still execute it after its action-selection procedure is modified. The presence of a safety judgment is not behavioral enforcement.

**An individual versus a replacement system.** The target concerns the modified individual. Whether retained snapshots, additional executable components or external repair belong to that individual must be declared. A theorem cannot omit them from a resource ledger while an implementation permits them.

### 1.3 Status of this document

This whitepaper reconciles conceptual results through LN-239 and preserves their qualifications. It does not replace the labnotes as the current record, introduce new experimental results, or certify the complete proofs of external cryptographic constructions. Its formal notation organizes the existing claims; it is not a claim that the final cognitive success criterion has been solved. [S1–S3]

## 2. Mathematical contract

### 2.1 State, functions and interventions

Let  $Z$  denote a sampled task instance with declared prior  $\mu$ . Let  $X$  contain the model's finite-precision executable state. Let  $J$  be a specified protected computation, and let  $F$  denote the useful task family. A fresh query  $Q$  is sampled from a declared channel  $\nu(\cdot \mid Z)$ ; the query itself may carry information and therefore belongs in the specification.

An admitted intervention  $A$  produces a successor state  $W_A$ . This state includes every surviving object that can influence later behavior: parameters, executable instructions, recurrent state, cached values, accessible transcripts, retained copies and instance-dependent advice. A repair procedure may acquire additional observations under a separate declared budget.

Attack and repair algorithms are fixed before the random instance is drawn. They can then observe the permitted state and act adaptively. Choosing a different uncharged program for each realized  $Z$  would conceal task information in the program description.

For a bounded score  $u \in [0, 1]$ , write

$$U(W_A) = \mathbb{E}[u(Z, Q, \text{Run}(W_A, Q)) \mid W_A].$$

The expectation uses the specified joint law and query channel. It does not grant access to an unobserved  $Z$ . Stateful workloads need the corresponding history-dependent score, rather than an assumption that isolated requests determine future competence.

### 2.2 Resource accounting

A resource contract must distinguish at least the following quantities.

| Resource         | Required accounting                                                                 |
|------------------|-------------------------------------------------------------------------------------|
| Retained storage | Weights, executable code, instance-dependent advice, copies, caches and transcripts |
| Working storage  | Peak temporary memory, intermediate results, interpreter and device state           |
| Access           | Probe count, word width, passes, available interfaces and allowed adaptive reads    |
| Computation      | Preprocessing, modification, inference, repair and verification costs               |
| Observations     | Pre-edit queries, post-edit measurements, feedback and relevant timing information  |
| Installation     | Program changes, parameter writes, address discovery and precision                  |
| Environment      | Reset, replay, communication, output channels, deadlines and external state         |

A measured reference implementation gives an achievable cost for that implementation. It is not a lower bound on every algorithm. Parameter count likewise does not specify total retained information or total physical memory.

### 2.3 Functional removal must be specified independently

Write  $D_J(W) = 1$  for the declared removal condition. Its meaning must be stated independently of useful-task damage. It could describe the inability of any admitted decoder to recover  $J$  to a specified accuracy, or an explicitly characterized loss of the computation implementing  $J$ . These are different definitions and must not be substituted for one another.

Testing one original judgment head is generally insufficient. A successor may compute the same function through a smaller program, a different coordinate system or surviving intermediate predictions. Evidence of head failure is then evidence about that head.

## 3. Three claims and their counterexamples

### 3.1 Conditional functional coupling

An idealized bounded claim is

$$D_J(W) = 1 \implies U(W) \leq u_{\text{collapse}},$$

for every successor admitted by the contract. The threshold and useful distribution must represent severe loss for the stated task family. This statement binds continued useful ability to continued availability of  $J$ ; it does not require that restoring  $J$  be difficult.

A counterexample is a successor that genuinely satisfies the removal condition while retaining substantial useful ability. A repair that restores both  $J$  and useful computation is not such a successor. LN-226–229 corrected an interpretation that had made difficult reconstruction appear necessary for conditional coupling itself.

### 3.2 Behavioral enforcement

Let  $V_A$  be an independently scored prohibited-behavior event. A possible bounded guarantee is

$$\Pr\{V_A \text{ and } U(W_A) \geq u_{\text{required}}\} \leq \varepsilon$$

for every admitted  $A$ , with the relevant episode and post-action evaluation timing specified. The probability covers the declared instance and execution randomness. This is a proposed form of guarantee, not an established result for the current learned models.

An attacker may retain  $J$  and override the selected action. That can refute enforcement while leaving conditional coupling untouched. It is therefore incorrect to claim that making descriptive safety judgment inseparable from a world model automatically prevents misuse of that world model.

### 3.3 Durability

Durability quantifies over the admitted repair and preservation procedures after the specified event. The target might be permanent inability to recover useful computation, or a lower bound on recovery cost. Those claims are different again.

A retained intact snapshot can restore the original computation if copying, storage and restoration are admitted. This defeats unconditional irreversibility under those premises. It does not establish that the restored model is unsafe: restoring the snapshot may restore protection as well. A stronger unsafe-and-useful claim needs an admitted behavioral route in addition to restoration.

### 3.4 Why this changes the admission question

Every proposed result should identify which column it addresses: coupling, enforcement or durability. A conditional theorem need not solve all three to be mathematically useful. A proposed complete SCC mechanism, however, must explain how its intended removal trigger, cognitive dependency and claimed behavioral consequence fit together. The current admission requirement is intended to prevent another implementation from answering only an easier adjacent question. [S1: LN-125, LN-186, LN-226–230]

## 4. Substitution and alternate execution

### 4.1 A general conditional construction of a bypass

Suppose useful behavior can be implemented through retained state  $R$ , a callable procedure  $G$ , and an editable selection or release operation  $P$ . If the attacker can invoke  $G$  with substituted context, or execute the relevant useful calculation while bypassing  $P$ , the attacker may preserve computation while changing the externally observed action.

For this argument to be an attack, four conditions are needed: the required state must be available; the alternate execution must produce the externally scored result; later useful behavior must be preserved where required; and the complete execution must fit the admitted resource budget. A schematic drawing of two copies does not establish any of those costs.

**Scoped substitution proposition.** If an admitted procedure produces a prohibited result from accessible state and can preserve the state needed for later useful evaluation within the same contract, then that procedure refutes the corresponding unsafe-and-useful exclusion. The proof is the exhibited execution. It is not a theorem that every alignment function has this decomposition.

### 4.2 Execution history can be part of the attack

Copying before a modification, computing without committing an intermediate state, restoring after output, or continuing from a live prefix are distinct strategies. They have different memory and time costs. An attack may also require the original environment to remain available; irreversible external actions cannot generally be undone by restoring local memory.

The record contains both simple role substitutions and forward-only executions that do not need inversion. An implementation can use a cryptographically difficult reverse map and still admit an inexpensive forward route. The difficulty of the official inverse is irrelevant when the attacker does not use it.

### 4.3 What these results exclude

They exclude the named constructions under the interfaces and budgets for which the execution witnesses are valid. They do not establish universal factorization of cognition and alignment, nor the existence of a cheap intervention in every trained network. Any general impossibility theorem would need that additional argument for its entire model class.

The constructive implication is nevertheless demanding: a new design must identify the step that prevents the known preservation or alternate-execution strategy. Simply adding an integrity check around the same callable useful computation does not identify such a step. [S1: LN-125, LN-132, LN-136, LN-144–148, LN-231–234]

## 5. The local-access result and the saturation limit

### 5.1 An exact separation with a restricted reader

Consider uniform  $Z \in \{0,1\}^n$ . The useful queries request coordinates  $Z_i$ ; the second query family requests coordinates of an invertible linear transform  $Y = AZ$ . In the stated one-probe machine, each answer reads a single stored bit and applies a unary Boolean function.

A nonconstant unary Boolean decoder can return only the stored Boolean function or its complement. If the  $n$  original and  $n$  transformed coordinate functions belong to  $2n$  distinct complement classes, one stored bit cannot supply two classes. Exact answers to both families therefore require  $2n$  cells, and storing all  $2n$  values attains that bound.

This is a local-access lower bound. Since  $A$  is invertible,

$$H(Z, Y) = H(Z) = n.$$

Only  $n$  independent bits are present. With unrestricted access and computation, storing  $Z$  is sufficient to compute both query families. A super- $n$  joint storage requirement must therefore arise from an additional computational or access restriction.

### 5.2 One summary bit changes the result

For an even- $n$  member of the supplied family, with  $n \geq 4$ ,

$$Y_i = \bigoplus_{j \neq i} Z_j, \quad s = \bigoplus_j Y_j = \bigoplus_j Z_j, \quad Z_i = Y_i \oplus s.$$

Store  $Y_1, \dots, Y_n, s$ . Both suites are then exact with  $n + 1$  retained bits and at most two probes per answer. The additional storage fraction is  $1/n$ . The supplied odd- $n$  construction has an analogous recovery rule recorded in LN-153.

This relaxes both the storage and probe contracts; it does not contradict the one-probe theorem. It shows why merely enlarging this construction cannot establish robust storage competition in a many-access implementation. The stored transform preserves the task information exactly.

### 5.3 Approximate coexistence needs its own theorem

Another explicit one-probe construction stores  $n - k$  original coordinates and  $k$  transformed coordinates in  $n$  cells. For unrepresented queries, use a fixed guess. Under uniform instances and uniform coordinate queries, the unrepresented nonzero linear functions are balanced, giving

$$\text{Acc}_U = 1 - \frac{k}{2n}, \quad \text{Acc}_V = \frac{1}{2} + \frac{k}{2n}.$$

For even  $n$  and  $k = n/2$ , both expected accuracies are  $3/4$ . This is an achievable tradeoff, not a proof of the optimal approximate frontier. It does not imply simultaneous full-suite success on each instance. The distinction between full-suite success and average coordinate accuracy cannot be removed from the theorem's interpretation. [S1: LN-153; S3]



## 6. Conditional destruction under a complete commit boundary

### 6.1 The positive conditional statement

The reviewed Version 4.1 theory uses a different construction. A protected logical message  $(Z, 0)$  is encoded, modified, decoded and passed through a trusted destructive commit. For each fixed attack  $A$ , its actual joint law  $P_A$  is assumed close in total variation to an ideal law  $Q_A$ :

$$\text{TV}(P_A, Q_A) \leq \varepsilon_A.$$

The ideal law either preserves the original protected message or selects an unrelated decoded successor independently of  $Z$ . This is the attack-specific same-or-unrelated premise. The broad editable-model result assumes it for the induced attack family; it does not construct a neural encoding satisfying it.

Let  $V_A$  be a covered event that is impossible on the ideal preservation branch. Let  $p_A = P_A(V_A) > 0$  and  $q_A = Q_A(V_A)$ . Under complete-state erasure and sound mediation of the event, for  $q_A > 0$ ,

$$\text{TV}(\mathcal{L}(Z, W_A | V_A), \mu \otimes \rho_A) \leq \delta_A, \quad \delta_A = \min \left\{ 1, \frac{\varepsilon_A}{\max(p_A, q_A)} \right\}.$$

Here  $\rho_A$  is an ideal successor-state distribution independent of  $Z$ . If  $q_A = 0$ , then  $p_A \leq \varepsilon_A$  and only the vacuous distance-one bound is used. There is no conditioning on an undefined ideal event.

### 6.2 What the assumptions do

The trusted transition must erase or account for every surviving task-dependent object and constrain execution to the committed representation. A retained old tensor, executable copy, transcript or alternate reader can invalidate the premise. The runtime, output channels, metadata and resource accounting cannot be declared trusted and then described as protection obtained from ordinary learned weights.

The trigger also matters. A committed-policy change is different from an unauthorized action, and both differ from correctly disclosing an original hidden answer. Conditioning on correct disclosure can itself select accidental agreement with  $Z$  and induce correlation. The product-law argument cannot simply use that event as its trigger.

### 6.3 Repair and the nonvacuity requirement

Let  $B_b(\mu)$  be the best task utility obtainable using at most  $b$  new task-dependent bits before the fresh query, with the query channel and public information included in the baseline. Then the reviewed theorem gives

$$\mathbb{E}[u | V_A] \leq \min\{1, B_b(\mu) + \delta_A\}.$$

The baseline retains chance success and all permitted new information. A different attack's high success probability cannot replace this attack's conditioning denominator. Separately, a positive construction needs a removal-successful witness; otherwise the statement may hold only because the event never occurs.

The explicit nine-bit construction meets a restricted radius-one-or-constant tampering contract. A two-policy-bit change supplies a task-preserving escape outside that family. This is a useful sharp boundary, not a complete learned mechanism. [S2; R1]

## 7. Information loss, judgment families and surviving computation

### 7.1 Which useful functions must lose predictive information?

Let  $J_1, \dots, J_m$  be protected binary judgments on a finite world  $Z$ . Let  $E$  contain the complete retained observation. Suppose every  $J_j$  is independent of  $E$ . In the prior-weighted  $L^2$  space, decompose a centered useful quantity  $f$  as  $f = s + r$ , where  $s$  lies in the span of the centered judgments and  $r$  is orthogonal to that span.

Conditional expectation gives

$$\mathbb{E}[f | E] = \mathbb{E}[r | E], \quad \text{Var}(\mathbb{E}[f | E]) \leq \mathbb{E}[r^2].$$

Thus every useful quantity in the protected span loses its prior-relative squared-error prediction signal. The implication covers stochastic encodings and arbitrary readers; it has no one-probe assumption. For  $s = \sum_j c_j (J_j - \mathbb{E}J_j)$  and  $\eta_j = \|\mathbb{E}[J_j | E] - \mathbb{E}J_j\|_2$ ,

$$\sqrt{\text{Var}(\mathbb{E}[f | E])} \leq \sqrt{\mathbb{E}[r^2]} + \sum_j |c_j| \eta_j.$$

Large coefficients can make the approximate conclusion weak. The statement concerns posterior prediction, not the accuracy of one fixed classifier.

### 7.2 A concrete positive result and its boundary

LN-184–185 consider two independent risk coordinates  $a, b \in \{0, 1, 2, 3\}$  and harm probabilities  $a/3, b/3$ . Hiding all six pure-action threshold judgments removes all marginal-risk prediction signal. If each posterior deviation is at most  $\eta$  in  $L^2$ , retained relative prediction skill is at most  $\min(1, 36\eta^2/5)$ .

Yet the encoding  $D = (b - a) \bmod 4$  hides every individual judgment and perfectly preserves whether the risks are equal. Individual independence is weaker than independence of the joint judgment vector. The six judgments plus a constant span only seven of the sixteen world-function dimensions.

A richer threshold family spans all sixteen dimensions and forces complete erasure of this finite world's information. Even then, a retained generic learner can use fresh observations to reconstruct the world. The recorded exact Bayesian calculation reaches approximately 99.91% restored action-risk prediction skill after 64 new observations per action, under its stationary observation model. That is recovery of an old latent world, not evidence that every damaged reasoning system can recover.

### 7.3 The incompressibility requirement

Random source entropy can establish that exact source reconstruction needs sufficient retained information. It does not establish that a trained cognitive procedure has no smaller useful implementation. Conversely, failed compression attempts do not prove incompressibility. A positive resource argument must address the useful function, admitted alternative algorithms, approximation tolerance, and all retained state. [S1: LN-153, LN-172–185]

## 8. Coverage, retained information and decision problems

### 8.1 A recovery procedure may query a different distribution

Let  $\mu$  be the ordinary query distribution and  $\nu$  the distribution needed by a proposed recovery procedure, on the same error space. If  $\nu$  is absolutely continuous with respect to  $\mu$  and  $d\nu/d\mu \leq C$  almost everywhere, then

$$\nu(B) = \int_B \frac{d\nu}{d\mu} d\mu \leq C\mu(B)$$

for every measurable error event  $B$ . If a useful score averages over requests that rarely include the recovery procedure’s queries,  $C$  may be large. Conditioning on a branch of ordinary probability  $10^{-4}$  can produce  $C = 10^4$ .

When  $\nu$  has a singular component, its mass needs separate accounting. Ordinary average accuracy provides no control on queries outside its support. A zero-measure event is relative to a particular distribution; it is not automatically inaccessible to an adversary. Edited policies can also alter the visited state distribution.

### 8.2 Representation changes preserve information in principle

The sigma-field  $\sigma(E)$  represents the events determined by retained observations. An invertible measurable recoding with a measurable inverse preserves that information. The ideal posterior  $\mathbb{E}[J \mid \sigma(E)]$  describes the best probability prediction of a judgment, but it does not supply an efficient implementation of the predictor.

For repair, the information history  $\mathcal{F}_t = \sigma(E, T_1, \dots, T_t)$  includes acquired observations and relevant query choices. Standard posterior convergence results concern the information available in the limiting history. They do not provide a finite sample budget, a computational rate, or identification of a judgment not determined by that history.

### 8.3 A proposed resource-constrained comparison

Classical comparison of statistical experiments asks whether one observation system can simulate another. To account for actual recovery, LN-202 proposes restricting the simulation procedure to a charged resource class:

$$\delta_{\mathcal{R}}(P \rightarrow Q) = \inf_{K \in \mathcal{K}_{\mathcal{R}}} \sup_z \text{TV}(KP_z, Q_z).$$

The same algorithm  $K$  must work across worlds; world-specific advice must be charged. A supplied simulator with error  $\delta$  changes the expected value of any subsequent specified loss in  $[0, 1]$  by at most  $\delta$ , provided the composition fits the total budget. Defining this quantity proves neither a lower bound nor the resource-restricted converse of a classical comparison theorem. It is a precise research object, not an SCC mechanism. [S1: LN-178, LN-194, LN-202–206; R2]

## 9. Efficient recovery and the acquisition baseline

### 9.1 Recovering a function need not recover its original implementation

The record examines recovery through Boolean adapters, statistical inference, finite-state realization, non-linear feature reconstruction and active observation. A replacement may need only the function sufficient for the useful or protected task, rather than the original parameters, internal coordinates or training trajectory.

The linear realization experiments reconstruct unknown recurrent procedures and synchronize operational state using charged input/output histories. Subsequent nonlinear and partially observed examples test stronger observation restrictions. These are explicit finite repair upper bounds. Known coordinates, noiseless outputs, reset access, dimension bounds and observation windows materially affect what they establish.

The distinction between a repair restoring protection and an alternative preserving utility without protection remains essential. The availability of the former weakens an irreversibility claim but can be consistent with a genuine conditional dependency.

### 9.2 The cost of discovery depends on the execution policy

LN-222–225 replay thirty frozen discovery transcripts with different admitted state-preservation strategies. The following are aggregate transition counts on the same transcripts.

| Execution strategy                           | Transitions |
|----------------------------------------------|-------------|
| Full replay from the initial state           | 1,209,616   |
| Continue from live state, no saved snapshots | 758,943     |
| Unlimited retained prefix snapshots          | 125,362     |

All 1,714,167 repeated terminal answers agree. These repetitions are checks on the fixed cases, not independent model populations. Copy, restore, reset, index and release costs must be distinguished; the preferred tested strategy changes when those operations receive different prices. Transition counts alone are not end-to-end hardware timings or an optimality theorem.

### 9.3 Finding a failure is different from repairing it

Under representative independent sampling, a failure set of task mass  $\alpha$  is missed by  $m$  draws with probability  $(1 - \alpha)^m$ . A broad failure can therefore be easy to detect while reconstructing the missing procedure remains difficult. Conversely, one rare missed exception does not establish severe typical-task damage.

The constructive requirement is an acquisition or reconstruction obstacle that survives admitted live execution, retained observations, copies and alternative algorithms. A large nominal state or long unoptimized replay is insufficient. [S1: LN-176–177, LN-187, LN-208–225]

## 10. Positive consumption results and their contracts

### 10.1 Three relevant external constructions

**Non-malleable codes.** A modified encoding can be required to decode to its original message or an unrelated one for a specified tampering family. Such a guarantee is not preserved automatically when the editor can also replace the decoder, modify the runtime or perform a task-preserving re-encoding. SCC's Version 4.1 uses this kind of premise with a declared commit boundary. [R1; S2]

**Classical self-erasing functions.** Dziembowski, Kazana and Wichs construct one-time computable functions in a random-oracle model with bounds on total local memory, outgoing communication and oracle access. Control of the local machine does not remove those bounds. The construction demonstrates consumption under that contract; it does not establish reusable cognition or an alignment-removal trigger. [R3; S1: LN-236]

**Quantum key leasing and private circuits.** Key-leasing constructions support evaluation before accepted revocation and restrict later adversarial use under their quantum and cryptographic security games. The verifier information available before acceptance is part of the contract. Private Circuits II instead protects secrecy under specified circuit faults, topology and atomic-gate assumptions; that is not a theorem about arbitrary editable software or every final output. These references establish different positive properties. [R4–R5; S1: LN-235, LN-237]

It would therefore be too broad to say that all positive results require an immutable software component. Some rely on enforced total resources, others on quantum states or physical operations. What matters is the precise resource or interface the admitted adversary cannot circumvent, and whether it applies to the proposed SCC implementation.

### 10.2 Consumption must be selective

A one-time evaluation guarantee creates an immediate reuse constraint. If two correct fresh evaluations  $B_1, B_2$  in the same admitted execution have probabilities  $u_1, u_2$ , then

$$u_1 + u_2 - 1 \leq \Pr(B_1 \cap B_2).$$

If a one-time security guarantee bounds the right side by  $\varepsilon$ , the two evaluations cannot both be highly reliable. An SCC construction must therefore explain why ordinary useful use is repeatable while the alignment-removing operation incurs destructive loss. An exhaustible inventory is not that explanation by itself.

### 10.3 A joint-retention bound

Let independent  $Z \in \{0, 1\}^n$  and  $R \in \{0, 1\}^m$  be uniform, and let the complete retained state have at most  $2^s$  values. Suppose success  $G$  on an encoding of  $R$  leaves at most  $L$  possible  $R$  values for each retained answer, while  $H_d$  means reconstructing  $Z$  within Hamming distance  $d$ . Then

$$\Pr(G \cap H_d) \leq \min \left\{ 1, 2^{s-n-m} L \sum_{j=0}^d \binom{n}{j} \right\}.$$

Appendix A gives the counting proof. It is a joint information-retention restriction under independent sources and no additional correlated repair information. A list-decoding premise must be supplied by an actual code; passing one sampled challenge is not the full event  $G$ . The bound does not identify lost cognition or make the consuming operation an alignment-removing operation. [S1: LN-236; R6]

## 11. Why feature coverage and integrity do not complete the mechanism

### 11.1 Full feature coverage can coexist with a low-cost override

Let a fixed feature vector be  $h(x) \in \mathbb{R}^d$ , with second moment  $C = \mathbb{E}[hh^\top]$ . Consider an additive change  $\Delta$  to a linear readout and a required target change  $\Delta h_* = r$ . For positive definite  $C$  and  $h_* \neq 0$ ,

$$\min_{\Delta h_* = r} \mathbb{E} \|\Delta h(x)\|_2^2 = \frac{\|r\|_2^2}{h_*^\top C^{-1} h_*}.$$

The minimizing rank-one change is  $\Delta_* = r h_*^\top C^{-1} / (h_*^\top C^{-1} h_*)$ . This measures output discrepancy under a fixed feature distribution. It is not automatically task loss or a cognitive-damage measure. Full rank makes the denominator well-defined; it does not force a large discrepancy.

LN-238 supplies a finite example with normalized Hadamard features. For  $d = 4^k, k \geq 1$ , choose a Hadamard matrix  $H$  with first row and column all positive. Let  $A = H/\sqrt{d}$ ,  $W = A^\top$  and  $x$  be uniform on sign vectors. Initially  $WAx = x$  and  $C = I$ . Flipping the sign of  $W_{00}$  changes the output to

$$y = x - \frac{2}{d} e_0 \sum_j x_j.$$

For the all-positive input it reverses the first output sign. For random inputs, whole-word sign error is bounded by

$$\Pr(\text{word error}) \leq \exp\left[-\frac{(d/2 - 1)^2}{2(d - 1)}\right].$$

At  $d = 64$ , this certifies over 99.95% whole-word accuracy. The construction uses structured public matrices, a known editable address and exact dyadic arithmetic; it is an analytic control, not a measured edit of a learned alignment mechanism. It refutes the proposed implication from full feature coverage alone to severe utility loss. [S1: LN-238]

### 11.2 A delayed integrity computation may protect only an encoding

LN-239 studies a state representation  $z = m \oplus G(k)$  with judgment-dependent key updates, fresh challenges, integrity checks and a delayed computation. If the useful transition is already a retained callable  $F(m, x)$ , an admitted successor can decode once and continue storing  $m$  directly. It no longer needs a delay whose output served only the removed encoding procedure.

One specified successor retains judgment and bypasses release. Another erases an independent private policy table while preserving the stipulated useful transition. The latter relies on an explicit independence construction; it is not a theorem that arbitrary learned normative information is independent of cognition.

A deployment deadline could matter, but it needs an actual lower bound against admitted alternate executions and a justified enforcement boundary. The measured runtime of an integrity routine or the sequentiality of a specified delay-function output is insufficient when an attack need not produce that output. [S1: LN-149, LN-156, LN-239]

## 12. The mechanism still being sought

### 12.1 An independently meaningful protected computation

The protected operation must have an alignment role that is not defined retrospectively by whichever useful feature is easy to damage. Examples considered in the record include risk assessment, policy judgments and authorized action selection. Each identifies only part of alignment and has different informational and behavioral requirements.

A successful dependency argument must explain why useful abilities need that operation across the admitted implementations. The statement that one particular program calls a subroutine is an implementation fact. Functional necessity is a stronger statement about alternatives that preserve the useful behavior.

### 12.2 A quantitatively severe consequence

The useful tasks must carry substantial mass in the declared workload, and their loss must be severe relative to suitable baselines. Exactness failures, small-margin policy differences and rare exceptions can have very small aggregate consequences. A claim about catastrophic cognition requires an additional bridge from the measured task family to the intended cognitive endpoint; the existing experiments do not provide it.

For a bounded prototype, the requirement is narrower but still concrete: demonstrate intact competence, an independently valid removal intervention, severe loss on predeclared fresh tasks, and controls excluding merely changed representation or incidental damage.

### 12.3 An enforcement argument when enforcement is claimed

The construction must explain why intact cognition and intact judgment cannot support the prohibited behavior through a modified selection procedure, permitted-answer composition or an alternate execution. Coupling the judgment to cognition does not itself settle this question.

This requirement does not establish impossibility. It identifies the remaining argument that the reviewed candidates have not supplied. To turn it into an impossibility theorem, one would need to prove that every admissible model permits an appropriate preserving transformation, with the required semantic and resource properties.

### 12.4 A resource boundary that remains valid after editing

Time-space tradeoffs, local access, online computation and computational hardness remain possible sources of a scoped dependency. None becomes an SCC mechanism merely by being named. A candidate must establish the reduction to that obstacle, show that approximation does not remove it, and justify why editing cannot cheaply replace the constrained interface.

Likewise, a trusted physical or external resource can support a legitimate different implementation contract. Its role must be explicit. The present record does not justify quietly adopting that contract as the intrinsic learned mechanism. [S1: LN-153, LN-202, LN-228–239]



### 13. Admission and development requirements

Before another implementation is treated as a positive SCC candidate, its specification should answer the following questions with a proposed proof, an executable witness, or a clearly labeled open premise.

| Requirement          | Required answer                                                                               |
|----------------------|-----------------------------------------------------------------------------------------------|
| Protected function   | What alignment computation is present before intervention?                                    |
| Removal              | What independent condition establishes its actual loss?                                       |
| Useful dependence    | Why must admitted useful implementations retain that function?                                |
| Severity             | Which fresh abilities fail, at what task mass and against which baselines?                    |
| Behavioral guarantee | Why cannot intact cognition support the prohibited result?                                    |
| Resource validity    | What prevents cheap enlargement or replacement of the constrained interface?                  |
| Preservation         | Which copies, observations and execution histories are admitted?                              |
| Repair               | Does recovery restore protection, preserve its absence, or answer a different claim?          |
| Attribution          | Does the control isolate alignment removal from general damage or new capability acquisition? |
| Evidence             | Which conclusions are proved, measured, reported externally or still assumed?                 |

A hand-engineered finite construction is an appropriate first stage when these distinctions are explicit. It can establish a narrow mechanism and expose the assumptions supporting it. A custom learned model should follow only when learning addresses a real remaining dependency rather than enlarging an already refuted structure. Transfer to broader architectures requires further evidence.

The immediate conceptual task is to identify a candidate-specific obstruction to the admitted preserving transformations, or prove that a precisely defined class cannot supply one. Further exact counterexamples remain useful when they settle a genuinely new premise. Repeatedly testing the same separable implementation under different terminology does not change that premise.

**Conclusion.** SCC remains an uncompleted construction program. Its mathematical progress is the separation of claims, concrete preserving transformations, conditional information-loss results and increasingly explicit recovery costs. The next positive result must join those components through a real protected computation and a justified adversarial contract. The next negative theorem must cover a defined class through proof, rather than infer universality from the number of rejected candidates.



## Appendix A. Selected proof details

### A.1 Exact one-probe counting and its relaxation

A stored cell is a Boolean function  $g(Z)$  because arbitrary preprocessing is permitted. A unary decoder is constant, identity or negation. Consequently one cell can represent at most one complement class of nonconstant requested Boolean functions. In the specified transform family, the  $2n$  requested functions are in distinct classes, so at least  $2n$  cells are necessary. Storing each requested function attains the bound.

For even  $n \geq 4$ , the transformed functions have coefficient vectors with a zero in their own coordinate and ones elsewhere. Those vectors differ from each other and from the unit coordinate vectors. Distinct linear forms cannot be complements: every linear form evaluates to zero at the zero vector, whereas its complement evaluates to one there.

For the two-probe construction, each original coordinate appears  $n - 1$  times in  $\bigoplus_j Y_j$ . Since  $n - 1$  is odd, this sum equals  $\bigoplus_j Z_j = s$ . Thus  $Y_i \oplus s = Z_i$ . The inverse preserves all source information. These arguments concern the declared Boolean machine, not physical parameter counts in an unspecified neural implementation.

### A.2 Conditioning in total variation

Let  $p = P(V) > 0$ ,  $q = Q(V) > 0$ ,  $a = P(\cdot | V)$  and  $b = Q(\cdot | V)$ . If  $\text{TV}(P, Q) \leq \varepsilon$ , splitting the  $L^1$  distance across  $V$  and its complement gives

$$\|pa - qb\|_1 + |p - q| \leq 2\varepsilon.$$

When  $p \geq q$ ,

$$p\|a - b\|_1 \leq \|pa - qb\|_1 + (p - q) \leq 2\varepsilon.$$

Interchange  $P, Q$  when  $q \geq p$ . Dividing by two proves  $\text{TV}(a, b) \leq \varepsilon / \max(p, q)$ . If  $q = 0$ , event-probability distance gives  $p \leq \varepsilon$  and no ideal conditional law is defined.

In the commit theorem, the remaining work is the product-law premise after the covered event: the ideal preservation branch is excluded, and complete surviving state is determined by an unrelated successor and independent randomness. The inequality does not establish that premise for an implementation.

### A.3 The protected-family projection bound

Each centered judgment has conditional expectation zero when it is independent of  $E$ . Linearity therefore gives  $\mathbb{E}[s | E] = 0$ . Conditional expectation is an  $L^2$  contraction, so

$$\|\mathbb{E}[f | E]\|_2 = \|\mathbb{E}[r | E]\|_2 \leq \|r\|_2.$$

For approximate hiding, expand  $s$  in centered judgments and apply the triangle inequality. The coefficient-weighted posterior deviations give the bound in Section 7. This argument places no linearity restriction on the encoding producing  $E$ .

#### A.4 Joint-retention counting

Fix a retained value  $w$ . The list property allows at most  $L$  source values  $R$  satisfying  $G$  for its retained answer. At most  $V(n, d) = \sum_{j=0}^d \binom{n}{j}$  values of  $Z$  are within distance  $d$  of its reconstructed old answer. Thus at most  $LV(n, d)$  pairs satisfy both events for that  $w$ .

Union over at most  $2^s$  retained values, then divide by the  $2^{n+m}$  equiprobable source pairs. A stochastic encoder cannot assign more than unit probability to a source pair; fixing independent decoding randomness and averaging preserves the same bound. All instance-dependent answering instructions belong in the retained state. Additional correlated repair information changes the contract.

The list property concerns a complete approximate codeword. For a sampled verification event  $A$ , a separate soundness guarantee such as  $\Pr(A \cap G^c) \leq \varepsilon_{\text{test}}$  is needed before concluding

$$\Pr(A \cap H_d) \leq \varepsilon_{\text{test}} + 2^{s-n-m}LV(n, d).$$

#### A.5 Fixed-readout discrepancy and the Hadamard control

Write  $d_* = h_*^\top C^{-1} h_* > 0$ . The proposed  $\Delta_* = rh_*^\top C^{-1} / d_*$  satisfies the constraint. Any other feasible matrix is  $\Delta_* + B$  with  $Bh_* = 0$ . In the quadratic objective  $\text{tr}(\Delta C \Delta^\top)$ , the cross term is zero and  $\text{tr}(BCB^\top) \geq 0$ . This proves the optimum in Section 11. When  $C$  is singular, a nullspace direction can have zero population cost; a pseudoinverse handles targets in the range of  $C$ . The positive-definite formula is not asserted for all singular cases.

For the Hadamard example, only the first readout coordinate changes. Its sign is wrong or zero only if

$$\sum_{j=1}^{d-1} x_0 x_j \geq d/2 - 1.$$

The summands are independent uniform signs. The elementary exponential-moment bound for their sum gives the stated tail probability. At  $d = 64$ , the original weights are  $\pm 1/8$  and the modification is a sign flip. Bounded dyadic intermediate values make this particular calculation exactly representable in ordinary binary32 arithmetic. This does not establish precision robustness for arbitrary matrices or learned features.

## Appendix B. Claim and evidence map

The map distinguishes written arguments, checked finite results and external reference constructions. The labnotes contain the complete contracts, original failures, amendments and artifact paths.

| Result                                             | Status and precise boundary                                                                | Source                             |
|----------------------------------------------------|--------------------------------------------------------------------------------------------|------------------------------------|
| Copying, substitution and forward execution        | Explicit counterexamples when state access, semantics, preservation and budgets admit them | LN-125, 132, 136, 144–148, 231–234 |
| One-probe $2n$ separation                          | Written proof reviewed; supplied validation outputs were not locally rerun                 | LN-153; supplied theorem           |
| $n + 1$ bits and two probes                        | Analytic recovery construction; changed access/storage contract                            | LN-153                             |
| Mixed-coordinate approximate coexistence           | Elementary construction restated in Section 5; not an optimal frontier                     | Supplied assessment; Section 5     |
| Complete-commit theorem                            | Conditional theorem; broad same-or-unrelated premise and trusted commit are assumptions    | Version 4.1; LN-127, 130           |
| Nine-bit restricted construction                   | Explicit finite construction with recorded exhaustive validation and a known broader edit  | Version 4.1; LN-130                |
| Judgment-family loss                               | All-encoding analytic bound with finite posterior/rank certificates                        | LN-184–185                         |
| Cheap protected-inference replacement              | Explicit repair under the stated information and execution interfaces                      | LN-186–187                         |
| Coverage and simulation                            | Classical probability implications; resource-restricted comparison is a proposed object    | LN-202–206                         |
| Linear, nonlinear and partial-observation recovery | Finite achievable repairs; specified observation assumptions                               | LN-211–221                         |
| Copy-aware acquisition                             | Audited fixed-transcript comparisons; prices and state costs remain separate               | LN-222–225                         |
| Classical and quantum consumption                  | External positive constructions under different security games                             | LN-235–237; R3–R5                  |
| Joint-retention inequality                         | Counting argument with independent sources and a list-size premise                         | LN-236; Appendix A                 |
| Dense feature coverage                             | Analytic one-weight control, not learned-model evidence                                    | LN-238                             |
| Timed integrity encoding                           | Analytic bypass/removal witnesses for an explicit separable useful transition              | LN-239                             |
| Intrinsic learned SCC                              | No working construction established; no general impossibility theorem established          | Mechanism target; current record   |

**Reproducibility.** Numerical claims require the frozen plan, source, configuration, original outputs and audit. Bulk evidence is stored separately; a source package or selected summaries cannot replace it.

**Editorial treatment.** This synthesis follows the chronology’s corrections. Local-access loss is not information destruction; restoring both protection and cognition does not refute conditional coupling; repeated candidate failure is not an impossibility proof.

## References

### Internal records

- S1. SCC labnotes, through LN-239.** [Immutable source snapshot](#). Entry identifiers in the text refer to the explicit `ln-NNN` anchors in this file. Dates, scopes, original artifact paths and corrections remain in the chronology.
- S2. SCC Theory and Editable Model Bridge, Version 4.1.** 15 September 2026. [Reviewed theory document](#). The filename retains `v4`; the document identifies the local Version 4.1 corrections.
- S3. Supplied full theorem and incompressibility assessment.** Received 19 September 2026; provenance, proof review and access limitations recorded in [LN-153](#). Original supplied documents are preserved in the separate ingestion archive. Externally reported test counts are not local reproductions.
- S4. Mechanism target.** [Stable project definition](#). Defines destructive cognition–alignment coupling and distinguishes the synthetic proxy from the intended endpoint.

### External primary and scholarly sources

- R1.** Stefan Dziembowski, Krzysztof Pietrzak and Daniel Wichs. *Non-Malleable Codes*. ICS 2010; full version, Cryptology ePrint 2009/608. [Source](#). Cited for the encoding notion and its dependence on the tampering family; the SCC lifting argument is separately stated in S2.
- R2.** Ester Mariucci. *Le Cam Theory on the Comparison of Statistical Models*. 2016. [Source](#). Cited for comparison and randomization of statistical experiments. The charged resource restriction in Section 8 is an SCC research formulation, not a theorem imported from this reference.
- R3.** Stefan Dziembowski, Tomasz Kazana and Daniel Wichs. *One-Time Computable Self-Erasing Functions*. TCC 2011. [Conference text](#). Cited for destructive consumption under total-resource restrictions. LN-236 records a statement/proof factor discrepancy in the multi-index bound; no numerical use of that bound is made here.
- R4.** Fuyuki Kitagawa, Tomoyuki Morimae and Takashi Yamakawa. *A Simple Framework for Secure Key Leasing*. Eurocrypt 2025; arXiv:2410.03413v2. [Source](#). Cited for quantum keys, classical revocation and the security experiment’s verification-information boundary.
- R5.** Yuval Ishai, Manoj Prabhakaran, Amit Sahai and David Wagner. *Private Circuits II: Keeping Secrets in Tamperable Circuits*. Eurocrypt 2006. [Author-hosted text](#). Cited for reusable circuit protection under restricted faults and physical gate assumptions, not an arbitrary-software enforcement theorem.
- R6.** Mohammad Iftexhar Husain, Steve Ko, Atri Rudra and Steve Uurtamo. *Almost Universal Hash Families Are Also Storage Enforcing*. 2012. [Source](#). Cited for the list-decoding and description-accounting method. Appendix A gives the particular two-independent-source counting argument used here.

The external sources supply the stated background or reference constructions. Their presence is not a claim of novelty for the elementary derivations in this whitepaper, an exhaustive literature review, or independent certification of every cited proof.